Chapter 6. Building and Applying Logistic Regression Models

Model selection, diagnostics, conditional inference

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6.1 Strategies in model selection

Motivation. With several explanatory variables the number of candidate logistic models explodes. We need a model complex enough to fit well but simple enough to interpret, and principled tools (tests, AIC, subject-matter reasoning) to choose among candidates.

Confirmatory studies compare a few models chosen in advance by the study questions; exploratory studies search more widely and treat the result as a source of hypotheses. In either case, first look at the marginal effect of each predictor (a smoothed plot for a continuous x, a contingency table for a categorical one).

Practical guidelines

(a) *Events per variable.* A rough rule asks for at least 10 outcomes of each type per predictor: if $y = 1$ only 30 times among $n = 1000$, use at most about three x terms.
(b) *Multicollinearity.* Strongly correlated predictors can make each one look unimportant given the others, even when the global test is highly significant. Dropping a redundant predictor reduces the standard errors of the others.

6.1.1 Horseshoe crab example revisited

Example 6.1 (Horseshoe crabs: all main effects)

For 173 female crabs, $y = 1$ if she has at least one satellite. Predictors: color C (4 categories), spine condition S (3), weight, and carapace width W. The main-effects model

$$\text{logit}\, P(Y = 1) = \alpha + \beta_1\, \text{weight} + \beta_2\, \text{width} + \beta_3 c_1 + \beta_4 c_2 + \beta_5 c_3 + \beta_6 s_1 + \beta_7 s_2$$

(dummies for the first three colors and first two spine conditions) gives the LR test of $H_0 : \beta_1 = \cdots = \beta_7 = 0$ equal to 40.6 ($df = 7$, $P < 0.0001$). Yet no single effect is clearly significant:

Parameter	Estimate	SE	Parameter	Estimate	SE
Intercept	−9.273	3.838	color 2	1.506	0.567
weight	0.826	0.704	color 3	1.120	0.593
width	0.263	0.195	spine 1	−0.400	0.503
color 1	1.609	0.936	spine 2	−0.496	0.629

A significant global test with insignificant individual effects is a warning sign of multicollinearity: weight and width have correlation 0.887. We keep width W and use color C and spine condition S as factors. Models are named by their highest-order terms: $C + S + W$ (main effects), $C + S{*}W$ (adds the $S \times W$ interaction), etc. A model with an interaction should also contain the corresponding main effects.

6.1.2 Stepwise procedures

Forward selection adds, at each stage, the term giving the greatest improvement (smallest P-value, since different terms have different df); the *stepwise* variant re-tests earlier terms at each stage. *Backward elimination* starts from a complex model and at each stage removes the term whose removal hurts least (largest P-value), stopping when every deletion significantly worsens the fit.

Cautions

(a) Add or drop a factor (or an interaction containing it) as a whole, never single dummy variables; otherwise the result depends on the coding.
(b) Backward elimination is usually preferred: forward selection may stop early because one test in the sequence has low power.
(c) With many tests, unimportant terms can look impressive by chance, and the largest sample effect tends to overestimate its true effect. If forward, backward and best-subset searches (or a search on a bootstrap sample) disagree, the selected model is of dubious value.
(d) A variable central to the study may be kept and reported even if not significant; it can reduce bias in other effects and make results comparable with other studies.

6.1.3 Backward elimination for the horseshoe crab example

The deviance G^2 against the saturated model is not approximately χ^2 with a continuous predictor, but differences of deviances between nested models differing by a few parameters are LR statistics $-2(L_0 - L_1)$ with approximate χ^2 null distributions.

Example 6.2 (Horseshoe crabs: backward elimination)

Selected rows of the book's Table 6.2 (C color, S spine, W width; "vs." gives the comparison and deviance difference):

	Model	G^2	df	AIC	Comparison	Difference
1	$C{*}S{*}W$	170.44	152	212.4	–	–
2	$C{*}S + C{*}W + S{*}W$	173.68	155	209.7	(2)–(1)	3.2 ($df = 3$)
3c	$C{*}S + C{*}W$	173.69	157	205.7	(3c)–(2)	0.0 ($df = 2$)
4b	$W + C{*}S$	177.61	160	203.6	(4b)–(3c)	3.9 ($df = 3$)
5	$C + S + W$	186.61	166	200.6	(5)–(4b)	9.0 ($df = 6$)
6c	$C + W$	187.46	168	197.5	(6c)–(5)	0.8 ($df = 2$)
7a	C	212.06	169	220.1	(7a)–(6c)	24.5 ($df = 1$)
7b	W	194.45	171	198.5	(7b)–(6c)	7.0 ($df = 3$)
8	$C = \text{dark} + W$	187.96	170	194.0	(8)–(6c)	0.5 ($df = 2$)
9	None	225.76	172	227.8	(9)–(8)	37.8 ($df = 2$)

Starting from $C{*}S{*}W$, the three-factor interaction is dropped ($P = 0.36$), then $S \times W$, then the remaining two-factor interactions ($P = 0.27$ and then $P = 0.17$). Among main effects S can go, but removing C from $C + W$ costs 7.0 on $df = 3$ ($P = 0.07$) and removing W costs 24.5. Replacing the four colors by a single dummy (dark vs. not) loses almost nothing. Final model: width plus a "dark" indicator.

6.1.4 AIC, model selection, and the "correct" model

No model is exactly true; a parsimonious model with small bias may estimate quantities of interest (e.g. cell probabilities) better than a complex one. The *Akaike information criterion* judges a model by how close its fitted values are expected to be to the truth.

Definition (AIC)

For a model with p parameters and maximized log likelihood L_M,

$$\text{AIC} = -2\,(L_M - p) = -2L_M + 2p.$$

Smaller is better: the first term rewards fit, the second penalizes complexity.

Proposition 6.3 (AIC and the adjusted deviance)

Let models for a categorical response be fitted to the same data, with N observations in grouped form (so the saturated model has N parameters and the residual degrees of freedom are $df = N - p$). Then

$$\text{AIC} = [G^2 - 2\,df] + \text{const},$$

where the constant $-2L_S + 2N$ is the same for all models. Hence ordering models by AIC is the same as ordering them by $G^2 - 2\,df$.

Proof. By definition $G^2 = -2(L_M - L_S)$, so $-2L_M = G^2 - 2L_S$. Substituting $p = N - df$,

$$\text{AIC} = G^2 - 2L_S + 2(N - df) = [G^2 - 2\,df] + (2N - 2L_S).$$

The last bracket depends only on the data and the saturated model. ■

Theorem 6.4 (Akaike's bias correction)

Let $Y_1, \dots, Y_n$ be i.i.d. with density $f(\cdot;\theta_0)$ belonging to a regular parametric model with p-dimensional parameter, $\hat\theta$ the MLE and $\ell(\theta) = \sum_i \log f(Y_i;\theta)$. Let $Y_1^*, \dots, Y_n^*$ be an independent copy of the data and $\ell^*(\theta)$ its log likelihood. Then, to first order,

$$E[\ell(\hat\theta)] - E[\ell^*(\hat\theta)] \approx p.$$

Thus $-2\ell(\hat\theta) + 2p = \text{AIC}$ is an approximately unbiased estimate of $-2E[\ell^*(\hat\theta)]$, the expected out-of-sample deviance (equivalently, of $2n$ times the Kullback–Leibler discrepancy of $f(\cdot;\hat\theta)$ from the truth, up to a constant).

Proof. [Supplement] Let I be the Fisher information per observation, so $-\ell''(\theta_0)/n \to I$ and $\sqrt{n}(\hat\theta - \theta_0) \xrightarrow{d} N(0, I^{-1})$.

Step 1. Expand ℓ about $\hat\theta$, where $\ell'(\hat\theta) = 0$:

$$\ell(\theta_0) \approx \ell(\hat\theta) - \tfrac{n}{2}(\hat\theta - \theta_0)^\top I (\hat\theta - \theta_0).$$

Since $n(\hat\theta - \theta_0)^\top I(\hat\theta - \theta_0) \xrightarrow{d} \chi^2_p$ with mean p, $E[\ell(\hat\theta)] \approx E[\ell(\theta_0)] + p/2$.

Step 2. Expand ℓ^* about θ_0. Because $\ell^{*\prime}(\theta_0)$ is independent of $\hat\theta$ with mean zero,

$$E[\ell^*(\hat\theta)] \approx E[\ell^*(\theta_0)] + E[\ell^{*\prime}(\theta_0)^\top(\hat\theta - \theta_0)] - \tfrac{n}{2}E[(\hat\theta - \theta_0)^\top I(\hat\theta - \theta_0)] = E[\ell^*(\theta_0)] - \tfrac{p}{2}.$$

Step 3. $\ell(\theta_0)$ and $\ell^*(\theta_0)$ have the same distribution, so subtracting the two displays gives $E[\ell(\hat\theta)] - E[\ell^*(\hat\theta)] \approx p$. Hence $E[-2\ell(\hat\theta) + 2p] \approx -2E[\ell^*(\hat\theta)]$. Finally $-2E[\ell^*(\theta)]/n = 2\,\mathrm{KL}(f_{\theta_0} \| f_\theta) - 2E\log f(Y;\theta_0)$, and the last term does not depend on the model. ■

Example 6.5 (Horseshoe crabs: AIC)

Among models built from C, S, W, AIC is smallest for $C + W$ (197.5); the dark-indicator model does better still (194.0). Its advantage should be weighed against the fact that it was suggested by the $C + W$ fit.

6.1.5 Using causal hypotheses to guide model building

A time ordering among variables suggests a sequence of logit models, each response being explained by variables earlier in the chain.

Example 6.6 (Divorce and pre/extramarital sex)

A British study cross-classified G = gender, P = premarital sex, E = extramarital sex, M = marital status:

	Women				Men			
PMS	Yes		No		Yes		No	
EMS	Yes	No	Yes	No	Yes	No	Yes	No
Divorced	17	54	36	214	28	60	17	68
Still married	4	25	4	322	11	42	4	130

The ordering $G \to P \to E \to M$ suggests a causal diagram in which G affects P; G, P affect E; and E, P (not G directly) affect M. Fitting each stage:

Response	Explanatory	G^2	*df*
P	none	75.3	1
E	none / P / $G+P$	48.9 / 2.9 / 0.0	3 / 2 / 1
M	$E+P$ / $E{*}P$ / $E{*}P+G$	18.2 / 5.2 / 0.7	5 / 4 / 3

Stage 1 shows a strong G–P association (odds ratio 0.27). At stage 2, adding G to P reduces G^2 by only 2.9, weak evidence of a direct G effect; the EP conditional odds ratio is 4.0. At stage 3, the hypothesized $E+P$ fits poorly; an $E \times P$ interaction is needed. The causal reasoning led close to, but not exactly to, a well-fitting model.

6.1.6 New model-building strategies for data mining

For huge data sets (*data mining*) almost any effect is significant, so tests are of little use; complex models may still be valuable for prediction. Nevertheless, beyond some point new predictors are nearly linear combinations of existing ones and add no predictive power. Summary measures of predictive power (Section 6.2) become more relevant than inference.

6.2 Logistic regression diagnostics

Motivation. A global goodness-of-fit statistic can hide where and why a model fails. Residuals and influence measures examine the fit observation by observation and suggest improvements.

6.2.1 Pearson, deviance, and standardized residuals

Let y_i be binomial with n_i trials at setting i ($i = 1, \ldots, N$) of the predictors, $\hat{\pi}_i$ the fitted probability, and $\hat{h}_i$ the leverage, the ith diagonal element of the estimated GLM hat matrix

$$\widehat{\text{Hat}} = \hat{W}^{1/2} X (X^\top \hat{W} X)^{-1} X^\top \hat{W}^{1/2}, \qquad \hat{W} = \text{diag}(n_i \hat{\pi}_i (1 - \hat{\pi}_i)).$$

Definition (Residuals for binomial GLMs)

(a) Pearson residual:

$$e_i = \frac{y_i - n_i\hat{\pi}_i}{\sqrt{n_i\hat{\pi}_i(1-\hat{\pi}_i)}}. \tag{6.1}$$

(b) Standardized (adjusted) Pearson residual: $r_i = e_i/\sqrt{1-\hat{h}_i}$.
(c) Deviance residual: $\sqrt{d_i}\ \operatorname{sgn}(y_i - n_i\hat{\pi}_i)$, where

$$d_i = 2\Big[y_i \log\frac{y_i}{n_i\hat{\pi}_i} + (n_i - y_i)\log\frac{n_i - y_i}{n_i - n_i\hat{\pi}_i}\Big].$$

By construction $X^2 = \sum_i e_i^2$ and $G^2 = \sum_i d_i$.

Theorem 6.7 (Approximate distribution of residuals)

Suppose the logistic model holds with p parameters and the n_i are large. Then, approximately,
(a) $\operatorname{Var}(y_i - n_i\hat{\pi}_i) \approx n_i\pi_i(1-\pi_i)(1-h_i)$, so $\operatorname{Var}(e_i) \approx 1 - h_i < 1$ while r_i is approximately $N(0,1)$;
(b) $E(\sum_i e_i^2) \approx N - p = df$, so $E(\sum_i e_i^2)/N \approx df/N < 1$.

Proof. [Supplement] Write $\mu = (n_i\pi_i)$, $W = \operatorname{diag}(n_i\pi_i(1-\pi_i)) = \operatorname{Cov}(y)$, $A = X^\top W X$.

Step 1. The likelihood equations for the canonical (logit) link are $X^\top(y - \hat{\mu}) = 0$. Since $\partial\mu/\partial\beta^\top = WX$, linearization gives $\hat{\mu} - \mu \approx WX(\hat{\beta} - \beta)$, and expanding the likelihood equations,

$$0 = X^\top(y - \hat{\mu}) \approx X^\top(y - \mu) - A(\hat{\beta} - \beta) \quad\Longrightarrow\quad \hat{\beta} - \beta \approx A^{-1}X^\top(y - \mu).$$

Step 2. Hence $y - \hat{\mu} \approx (I - WXA^{-1}X^\top)(y - \mu)$ and

$$\begin{aligned}\operatorname{Cov}(y - \hat{\mu}) &\approx (I - WXA^{-1}X^\top)\,W\,(I - XA^{-1}X^\top W)\\ &= W - 2WXA^{-1}X^\top W + WXA^{-1}(X^\top W X)A^{-1}X^\top W = W - WXA^{-1}X^\top W.\end{aligned}$$

Step 3. Pre- and post-multiplying by $W^{-1/2}$: $\operatorname{Cov}(W^{-1/2}(y - \hat{\mu})) \approx I - H$ with $H = W^{1/2}XA^{-1}X^\top W^{1/2}$, whose diagonal is (h_i). The ith component of $W^{-1/2}(y - \hat{\mu})$ is e_i (with π_i in place of $\hat{\pi}_i$, a negligible change), proving (a); asymptotic normality of e_i follows from that of y, and dividing by $\sqrt{1-h_i}$ gives unit variance.

Step 4. H is a projection of rank p, so $E\sum e_i^2 \approx \operatorname{tr}(I - H) = N - p$. ∎

When residuals are uninformative

With continuous predictors usually $n_i = 1$; then $y_i \in \{0,1\}$, e_i takes only two values, residual plots show two parallel bands, and the deviance itself carries no information about fit. Group observations with common predictor values whenever possible. With very small fitted counts, residuals lose relevance just as X^2 and G^2 do.

6.2.2 Heart disease example

Example 6.8 (Blood pressure and heart disease)

Framingham men aged 40–59, classified by blood pressure; response: coronary heart disease within six years. Scores x_i: $111.5, 121.5, \ldots, 161.5, 176.5, 191.5$.

Blood	Sample	Heart	Fitted		Std. residual	
pressure	size	disease	Indep.	Linear	Indep.	Linear
< 117	156	3	10.8	5.2	−2.62	−1.11
117–126	252	17	17.4	10.6	−0.12	2.37
127–136	284	12	19.7	15.1	−2.02	−0.95
137–146	271	16	18.8	18.1	−0.74	−0.57
147–156	139	12	9.6	11.6	0.84	0.13
157–166	85	8	5.9	8.9	0.93	−0.33
167–186	99	16	6.9	14.2	3.76	0.65
> 186	43	8	3.0	8.4	3.07	−0.18

The independence model $\text{logit}\,\pi_i = \alpha$ fits poorly ($G^2 = 30.0$, $X^2 = 33.4$, $df = 7$) and its residuals increase with x, suggesting the linear logit model $\text{logit}\,\pi_i = \alpha + \beta x_i$. For that model the trend disappears and $G^2 = 5.9$, $X^2 = 6.3$ ($df = 6$). Only category 2 has a largish residual ($r_2 = 2.37$; Pearson 2.01, deviance 1.85), unsurprising among eight residuals.

6.2.3 Graduate admissions example

Example 6.9 (Admissions at the University of Florida)

Applications to 23 departments, by gender G and admission A; π_{ik} = admission probability for gender i in department k. The model with no gender effect, $\text{logit}\,\pi_{ik} = \alpha + \beta_k^D$, fits rather poorly ($G^2 = 44.7$, $X^2 = 40.9$, $df = 23$). Admitted/rejected counts and the standardized residual for (female, admitted):

	Female		Male				Female		Male		
Dept	Yes	No	Yes	No	r	Dept	Yes	No	Yes	No	r
anth	32	81	21	41	−0.76	ling	21	10	7	8	1.37
astr	6	0	3	8	2.87	math	25	18	31	37	1.29
chem	12	43	34	110	−0.27	phil	3	0	9	6	1.34
clas	3	1	4	0	−1.07	phys	10	11	25	53	1.32
comm	52	149	5	10	−0.63	poli	25	34	39	49	−0.23
comp	8	7	6	12	1.16	psyc	2	123	4	41	−2.27
engl	35	100	30	112	0.94	reli	3	3	0	2	1.26
geog	9	1	11	11	2.17	roma	29	13	6	3	0.14
geol	6	3	15	6	−0.26	soci	16	33	7	17	0.30
germ	17	0	4	1	1.89	stat	23	9	36	14	−0.01
hist	9	9	21	19	−0.18	zool	4	62	10	54	−1.76
lati	26	7	25	16	1.65						

Astronomy and geography admitted significantly more females than predicted, psychology fewer; without these three departments the model fits well ($G^2 = 24.4$, $df = 20$). Adding a gender effect barely helps ($G^2 = 42.4$, $df = 22$), because the deviations go in different directions. That model's conditional GA odds ratio is 1.19 (odds of admission 19% higher for females given department), whereas the marginal odds ratio is 0.94: Simpson's paradox.

Proposition 6.10 (Residuals under conditional independence)

In a $2 \times 2 \times K$ table with $y_{ik} \sim \text{bin}(n_{ik}, \pi_{ik})$, the model $\text{logit}\,\pi_{ik} = \alpha + \beta_k$ has $\hat{\pi}_{1k} = \hat{\pi}_{2k} = y_{+k}/n_{+k}$, and in each stratum the raw residuals satisfy $y_{1k} - n_{1k}\hat{\pi}_{1k} = -(y_{2k} - n_{2k}\hat{\pi}_{2k})$. Consequently the two standardized residuals in stratum k are equal in magnitude and opposite in sign, whereas the two ordinary Pearson residuals need not be.

Proof.

Step 1. The model is saturated for the column $(y_{1k} + y_{2k})$ in each stratum and has no gender term, so the likelihood equations $\sum_i y_{ik} = \sum_i n_{ik}\hat{\pi}_{ik}$ with $\hat{\pi}_{1k} = \hat{\pi}_{2k}$ give $\hat{\pi}_{ik} = y_{+k}/n_{+k}$.

Step 2. Then

$$y_{1k} - n_{1k}\hat{\pi}_{1k} = y_{1k} - \frac{n_{1k}(y_{1k} + y_{2k})}{n_{+k}} = \frac{n_{2k}y_{1k} - n_{1k}y_{2k}}{n_{+k}} = -(y_{2k} - n_{2k}\hat{\pi}_{2k}).$$

Step 3. Two random variables that are negatives of each other have the same (estimated) standard error, so after standardizing by the true standard error of the raw residual the two residuals are $\pm$ the same number. The ordinary Pearson residuals divide instead by $\sqrt{n_{ik}\hat{\pi}_{ik}(1 - \hat{\pi}_{ik})}$, which differs for $i = 1, 2$ when $n_{1k} \neq n_{2k}$. ■

6.2.4 Influence diagnostics for logistic regression

Deleting a poorly fitted observation and refitting is equivalent to adding a parameter that fits it perfectly. High leverage means high potential influence, but in logistic regression the leverage depends on the fit through $\hat{W}$: an extreme x_i with $\hat{\pi}_i$ near 0 or 1 can have small leverage. Influence measures for observation i (a binary response or a binomial count) include:

(1) *Dfbeta*: the change in each $\hat{\beta}_j$ when i is deleted, divided by its SE;

(2) *confidence interval displacement* c_i: the change in a joint confidence region for β;

(3) ΔX_i^2, ΔG_i^2: the decrease in the fit statistics when i is deleted.

Theorem 6.11 (One-step influence approximations)

Let $A = X^\top \hat{W} X$, $w_i = n_i\hat{\pi}_i(1 - \hat{\pi}_i)$, $h_i = w_i x_i^\top A^{-1} x_i$, and let $\hat{\beta}_{(i)}$ be the one-step Newton approximation (started from $\hat{\beta}$) to the estimate without observation i. Then

$$\hat{\beta} - \hat{\beta}_{(i)} = \frac{A^{-1}x_i\,(y_i - n_i\hat{\pi}_i)}{1 - h_i}, \qquad c_i = (\hat{\beta} - \hat{\beta}_{(i)})^\top A(\hat{\beta} - \hat{\beta}_{(i)}) = \frac{e_i^2 h_i}{(1 - h_i)^2},$$

and the score statistic for the extra parameter that fits observation i perfectly equals $r_i^2 = e_i^2/(1 - h_i)$, the usual approximation to ΔX_i^2.

Proof. [Supplement]

Step 1. At $\hat{\beta}$ the full score is zero: $\sum_j x_j(y_j - n_j\hat{\pi}_j) = 0$. Hence the score of the reduced data is $U_{(i)}(\hat{\beta}) = -x_i(y_i - n_i\hat{\pi}_i)$, and its information is $A - w_i x_i x_i^\top$. One Newton step gives $\hat{\beta}_{(i)} = \hat{\beta} + (A - w_i x_i x_i^\top)^{-1} U_{(i)}(\hat{\beta})$.

Step 2. By the Sherman–Morrison formula,

$$(A - w_i x_i x_i^\top)^{-1} x_i = A^{-1}x_i + \frac{w_i A^{-1} x_i x_i^\top A^{-1} x_i}{1 - w_i x_i^\top A^{-1} x_i} = \frac{A^{-1}x_i}{1 - h_i},$$

which yields the formula for $\hat{\beta} - \hat{\beta}_{(i)}$.

Step 3. Then $c_i = (y_i - n_i\hat{\pi}_i)^2 x_i^\top A^{-1} x_i/(1-h_i)^2 = (y_i - n_i\hat{\pi}_i)^2 h_i/\{w_i(1-h_i)^2\} = e_i^2 h_i/(1-h_i)^2$.

Step 4. Add a parameter γ with covariate $z = \mathbf{e}_i$ (indicator of observation i). At $(\hat{\beta}, \gamma = 0)$ the score for γ is $U_\gamma = y_i - n_i\hat{\pi}_i$, and the information blocks are $I_{\gamma\gamma} = w_i$, $I_{\gamma\beta} = w_i x_i^\top$, $I_{\beta\beta} = A$. The efficient information for γ is $w_i - w_i^2 x_i^\top A^{-1} x_i = w_i(1-h_i)$, so the score statistic is $(y_i - n_i\hat{\pi}_i)^2/\{w_i(1-h_i)\} = r_i^2$. Fitting γ is the same as deleting observation i, so this approximates the resulting drop in X^2 (and G^2). ■

Example 6.12 (Heart disease: influence)

For the linear logit model of Example 6.8, the second category has by far the largest Dfbeta (-1.14), c (2.26), ΔX^2 (5.64) and ΔG^2 (5.04); all others are below 1.4. For the independence model, deleting an extreme category changes X^2 by up to 14.2, but that removes a whole binomial sample, not one subject.

6.2.5 Summarizing predictive power: R and R-squared measures

No analog of R^2 is as universally useful for categorical responses as R^2 is for linear models. Three proposals:

(a) The correlation $r(y, \hat{\mu})$ between observations and fitted values (for binary data, $\hat{\mu}_i = \hat{\pi}_i$). For the crabs, $r = 0.402$ for W, 0.452 for $C + W$, 0.447 for dark+W, 0.526 for $C{*}S{*}W$; the last is inflated by its many parameters (a jackknife adjustment removes most of the difference).

(b) Proportional reduction in squared error $1 - \sum(y_i - \hat{\pi}_i)^2/\sum(y_i - \bar{y})^2$. Neither (a) nor (b) need be monotone in model complexity for logistic regression.

(c) Likelihood-based measures, below.

Proposition 6.13 (Likelihood R^2 measures)

Let $L_0 \le L_M \le L_S$ be the maximized log likelihoods of the null (intercept-only), the given and the saturated models. Then $(L_M - L_0)/(L_S - L_0) \in [0, 1]$, and:
(a) for n ungrouped Bernoulli observations, $L_S = 0$ and the measure reduces to McFadden's $D = (L_0 - L_M)/L_0$, where $L_0 = n[\bar{y}\log\bar{y} + (1-\bar{y})\log(1-\bar{y})]$;
(b) for N grouped binomial counts it equals $D^* = \{G^2(0) - G^2(M)\}/G^2(0)$.

Proof. The parameter spaces are nested (null $\subset M \subset$ saturated), so $L_0 \le L_M \le L_S$, and all are ≤ 0 since probabilities are at most 1; this gives the range.

(a) For Bernoulli data the saturated model sets $\hat{\pi}_i = y_i \in \{0, 1\}$, so each term $y_i \log\hat{\pi}_i + (1 - y_i)\log(1-\hat{\pi}_i) = 0$ and $L_S = 0$. The null model has $\hat{\pi}_i = \bar{y}$, giving L_0.

(b) $G^2(M) = -2(L_M - L_S)$ and $G^2(0) = -2(L_0 - L_S)$, so $(L_M - L_0)/(L_S - L_0) = \{G^2(0) - G^2(M)\}/G^2(0)$. ■

Why D^* is misleading

$D^* = 1$ for any perfectly fitting model, however poorly individual responses can be predicted. If the model holds but the null model does not, as $n \to \infty$ with N fixed $G^2(M)$ stays $O_p(1)$ while $G^2(0) \to \infty$, so $D^* \to 1$. It confounds goodness of fit with predictive power; use D on ungrouped data.

6.2.6 Summarizing predictive power: classification tables and ROC curves

A *classification table* cross-classifies y with the prediction $\hat{y} = 1$ if $\hat{\pi}_i > \pi_0$ (often $\pi_0 = 0.5$), summarized by sensitivity $P(\hat{y} = 1 \mid y = 1)$ and specificity $P(\hat{y} = 0 \mid y = 0)$. It discards information, depends on the arbitrary π_0, and is sensitive to the relative frequencies of the outcomes.

Definition (ROC curve and concordance index)

The *ROC curve* plots sensitivity against 1−specificity as π_0 ranges over $[0, 1]$; it joins $(0, 0)$ to $(1, 1)$. The *concordance index* is

$$c = \frac{1}{n_1 n_0} \sum_{i:\, y_i = 1} \sum_{j:\, y_j = 0} \Big[\mathbb{1}\{\hat{\pi}_i > \hat{\pi}_j\} + \tfrac{1}{2} \mathbb{1}\{\hat{\pi}_i = \hat{\pi}_j\} \Big],$$

the proportion of (case, non-case) pairs in which the case has the larger fitted probability (n_1, n_0 = numbers of 1's and 0's).

Proposition 6.14 (Area under the ROC curve)

The area under the empirical ROC curve (with linear interpolation across ties) equals the concordance index c.

Proof. [Supplement]

Step 1. Lower the cutoff π_0 from 1 to 0. Each time it passes a distinct fitted value v, the curve moves from (a, b) by $(m_0(v)/n_0,\, m_1(v)/n_1)$, where $m_1(v), m_0(v)$ count cases and non-cases with $\hat{\pi} = v$. At that moment $b = \#\{i : y_i = 1, \hat{\pi}_i > v\}/n_1$.

Step 2. The area added by this segment is a trapezoid:

$$\frac{m_0(v)}{n_0} \Big(b + \frac{m_1(v)}{2n_1} \Big) = \frac{1}{n_0 n_1} \sum_{j:\, y_j = 0,\ \hat{\pi}_j = v} \Big[\#\{i : y_i = 1, \hat{\pi}_i > v\} + \tfrac{1}{2} \#\{i : y_i = 1, \hat{\pi}_i = v\} \Big].$$

Step 3. Summing over all distinct v runs over every non-case j exactly once, giving the double sum defining c. ■

A value $c = 0.5$ corresponds to random guessing (the diagonal ROC line of an intercept-only model). For the crabs, $c = 0.639$ (color), 0.742 (width), 0.771 (width and color), 0.772 (width and dark indicator).

6.3 Inference about conditional associations in $2 \times 2 \times K$ tables

Motivation. Comparing two treatments on a binary outcome while controlling for strata (centers, age groups, studies in a meta-analysis) is perhaps the most common application. Logit models, the Cochran–Mantel–Haenszel test and the Mantel–Haenszel estimator are the standard tools, and they turn out to be closely linked.

Example 6.15 (Eight-center clinical trial)

Active drug vs. control cream for curing an infection, at eight centers (μ_{11k}, $\mathrm{Var}(n_{11k})$ are null hypergeometric moments of the number of drug successes):

	Drug		Control				
Center	Succ.	Fail.	Succ.	Fail.	Odds ratio	μ_{11k}	$\mathrm{Var}(n_{11k})$
1	11	25	10	27	1.19	10.36	3.79
2	16	4	22	10	1.82	14.62	2.47
3	14	5	7	12	4.80	10.50	2.41
4	2	14	1	16	2.29	1.45	0.70
5	6	11	0	12	∞	3.52	1.20
6	1	10	0	10	∞	0.52	0.25
7	1	4	1	8	2.0	0.71	0.42
8	4	2	6	1	0.33	4.62	0.62

6.3.1 Using logit models to test conditional independence

For binary Y, binary X ($x_1 = 1$, $x_2 = 0$) and a categorical control Z, let $\pi_{ik} = P(Y = 1 \mid X = i, Z = k)$ and consider

$$\operatorname{logit} \pi_{ik} = \alpha + \beta x_i + \beta_k^Z, \qquad i = 1, 2,\ k = 1, \ldots, K. \tag{6.2}$$

It assumes a common conditional odds ratio $\theta_{XY(k)} = e^{\beta}$. Conditional independence is $H_0 : \beta = 0$, tested by the Wald statistic $(\hat{\beta}/\mathrm{SE})^2$ or the LR statistic comparing (6.2) with

$$\operatorname{logit} \pi_{ik} = \alpha + \beta_k^Z, \tag{6.3}$$

both with $df = 1$. Alternatively, a goodness-of-fit test of (6.3) against the saturated model has $df = K$; it is less powerful when the association is similar across strata (Section 6.4), but can be more powerful when its direction varies.

6.3.2 Cochran–Mantel–Haenszel test of conditional independence

Conditioning on both margins in each partial table, the count n_{11k} determines the table.

Proposition 6.16 (Null hypergeometric moments)

Under conditional independence, given the margins of partial table k, n_{11k} is hypergeometric with

$$\mu_{11k} = E(n_{11k}) = \frac{n_{1+k} n_{+1k}}{n_{++k}}, \qquad \mathrm{Var}(n_{11k}) = \frac{n_{1+k} n_{2+k} n_{+1k} n_{+2k}}{n_{++k}^2 (n_{++k} - 1)}.$$

Proof. [Supplement] Drop the index k and write $n = n_{++}$. Under H_0 and fixed margins, the n_{+1} successes are a random subset of the n subjects. Let $I_s = 1$ if subject s of row 1 is a success ($s = 1, \ldots, n_{1+}$), so $n_{11} = \sum_s I_s$, $P(I_s = 1) = n_{+1}/n$ and, for $s \neq t$, $P(I_s = I_t = 1) = n_{+1}(n_{+1} - 1)/\{n(n-1)\}$.

Step 1. $E(n_{11}) = n_{1+} n_{+1}/n$.

Step 2. With $q = n_{+1}/n$, $\mathrm{Var}(I_s) = q(1-q)$ and $\mathrm{Cov}(I_s, I_t) = \frac{n_{+1}(n_{+1}-1)}{n(n-1)} - q^2 = -\frac{q(1-q)}{n-1}$. Hence

$$\mathrm{Var}(n_{11}) = n_{1+} q(1-q) - n_{1+}(n_{1+} - 1)\frac{q(1-q)}{n-1} = n_{1+} q(1-q)\frac{n - n_{1+}}{n-1} = \frac{n_{1+} n_{2+} n_{+1} n_{+2}}{n^2 (n-1)}.$$

■

Definition (CMH statistic)

$$\text{CMH} = \frac{\left[\sum_k (n_{11k} - \mu_{11k})\right]^2}{\sum_k \text{Var}(n_{11k})}, \tag{6.4}$$

with an approximate χ^2_1 null distribution. Cochran's version uses the binomial-based variance $n_{1+k}n_{2+k}n_{+1k}n_{+2k}/n^3_{++k}$ instead.

When $\theta_{XY(k)} > 1$ in every stratum (or < 1 in every stratum), the deviations $n_{11k} - \mu_{11k}$ tend to share a sign and accumulate; the test works best when the association is similar across strata. The hypergeometric formulation also covers retrospective studies (column margins fixed) and randomized trials (randomization distribution). A continuity correction brings the P-value closer to the exact test (Section 6.7.5) but tends to be conservative.

Theorem 6.17 (CMH as a score test for the logit model)

For model (6.2) with independent binomial rows:
(a) the (unconditional) score statistic for $H_0 : \beta = 0$ is Cochran's statistic;
(b) the score statistic for β based on the conditional likelihood given all row and column margins is exactly CMH. In particular CMH and Cochran differ only by the factors $(n_{++k} - 1)/n_{++k}$ in the variance and are asymptotically equivalent to the Wald and LR tests of $\beta = 0$.

Proof. [Supplement] Absorb α into the stratum parameters: $\text{logit}\,\pi_{ik} = \beta x_i + \gamma_k$.

Step 1. (Score.) The log likelihood is $\sum_{ik}[y_{ik}(\beta x_i + \gamma_k) - n_{ik}\log(1 + e^{\beta x_i + \gamma_k})]$, so

$$\frac{\partial L}{\partial \beta} = \sum_k (y_{1k} - n_{1k}\pi_{1k}), \qquad \frac{\partial L}{\partial \gamma_k} = \sum_i (y_{ik} - n_{ik}\pi_{ik}).$$

Under H_0, $\hat\pi_{1k} = \hat\pi_{2k} = \hat\pi_k = n_{+1k}/n_{++k}$ (here $y_{ik} = n_{i1k}$), and the β-score becomes $U = \sum_k (n_{11k} - n_{1+k}n_{+1k}/n_{++k}) = \sum_k (n_{11k} - \mu_{11k})$.

Step 2. (Information.) With $v_k = \pi_k(1 - \pi_k)$ under H_0: $I_{\beta\beta} = \sum_k n_{1+k}v_k$, $I_{\beta\gamma_k} = n_{1+k}v_k$, $I_{\gamma_k\gamma_k} = n_{++k}v_k$, and $I_{\gamma_k\gamma_l} = 0$ for $k \neq l$. The efficient information for β is

$$I_{\beta\beta} - \sum_k \frac{I^2_{\beta\gamma_k}}{I_{\gamma_k\gamma_k}} = \sum_k v_k\Big(n_{1+k} - \frac{n^2_{1+k}}{n_{++k}}\Big) = \sum_k v_k \frac{n_{1+k}n_{2+k}}{n_{++k}}.$$

Plugging in $\hat v_k = n_{+1k}n_{+2k}/n^2_{++k}$ gives $\sum_k n_{1+k}n_{2+k}n_{+1k}n_{+2k}/n^3_{++k}$, so U^2 divided by this is Cochran's statistic, proving (a).

Step 3. (Conditional score.) Given all margins, the conditional distribution of $(n_{111}, \ldots, n_{11K})$ is the product of noncentral hypergeometric laws (Theorem 6.41 below):

$$P(n_{11k} = t_k,\ k = 1, \ldots, K) = \prod_k \frac{c_k(t_k)e^{\beta t_k}}{\sum_u c_k(u)e^{\beta u}},$$

a one-parameter exponential family in β with sufficient statistic $T = \sum_k n_{11k}$. Hence $\partial \log P/\partial\beta = T - E_\beta(T)$ and the conditional information is $\text{Var}_\beta(T)$. At $\beta = 0$ these are $\sum_k (n_{11k} - \mu_{11k})$ and $\sum_k \text{Var}(n_{11k})$ with the hypergeometric variance of Proposition 6.16, so the conditional score statistic is CMH, proving (b). The final claim follows from standard asymptotics for score, Wald and LR tests. ∎

6.3.3 Multicenter clinical trial example

Example 6.18 (Clinical trial: tests)

In Example 6.15 every center except the last shows a positive association, so combining is sensible: CMH = 6.38 ($df = 1$, $P = 0.012$). Model (6.2) gives $\hat{\beta} = 0.777$ (SE 0.307), Wald 6.42 ($P = 0.011$), LR 6.67 ($P = 0.010$).

6.3.4 CMH test and sparse data*

Standard ML asymptotics need K fixed as $n \to \infty$. The CMH test also works under *sparse-data asymptotics*, $K \to \infty$ with n, because its numerator and denominator are sums of K independent bounded terms. The extreme case is one matched pair per stratum ($n_{1+k} = n_{2+k} = 1$).

Proposition 6.19 (CMH for matched pairs is McNemar's test)

Suppose each stratum is a matched pair. Let b be the number of pairs in which only the first member succeeds and c the number in which only the second does. Concordant pairs contribute nothing to CMH, and

$$\text{CMH} = \frac{(b-c)^2}{b+c}.$$

Proof.

Step 1. If both members respond alike, one column total is 0, the table is determined by its margins, $n_{11k} = \mu_{11k}$ and $\text{Var}(n_{11k}) = 0$.

Step 2. For a discordant pair, $n_{+1k} = n_{+2k} = 1$, so $\mu_{11k} = 1/2$ and $\text{Var}(n_{11k}) = 1{\cdot}1{\cdot}1{\cdot}1/(2^2{\cdot}1) = 1/4$; $n_{11k} = 1$ in the b pairs and 0 in the c pairs.

Step 3. Hence $\sum_k (n_{11k} - \mu_{11k}) = b - (b+c)/2 = (b-c)/2$ and $\sum_k \text{Var} = (b+c)/4$, giving $(b-c)^2/(b+c)$. ∎

Even though each n_{11k} takes only two values, the CLT applies to the sum over the $K^* = b + c$ discordant pairs, so CMH is approximately χ^2_1. LR and Wald tests for (6.2) are trustworthy only when strata totals exceed roughly 5–10 and K is small relative to n.

6.3.5 Estimation of common odds ratio

Under (6.2) the common odds ratio is $\theta = e^{\beta}$, with ML estimate $e^{\hat{\beta}}$. Non-model-based alternatives are Woolf's weighted average of the log odds ratios and the Mantel–Haenszel estimator.

Definition (Mantel–Haenszel estimator)

$$\hat{\theta}_{MH} = \frac{\sum_k n_{11k} n_{22k}/n_{++k}}{\sum_k n_{12k} n_{21k}/n_{++k}} = \frac{\sum_k R_k}{\sum_k S_k} = \frac{R}{S}. \tag{6.5}$$

Theorem 6.20 (Unbiased estimating equation for $\hat{\theta}_{MH}$)

If the rows of each partial table are independent binomials and the odds ratio equals θ in every stratum, then $E(R - \theta S) = 0$. Consequently $\hat{\theta}_{MH}$ is consistent both as $n \to \infty$ with K fixed and as $K \to \infty$ with bounded strata sizes.

Proof. [Supplement]

Step 1. In stratum k, $n_{11k} \sim \text{bin}(n_{1+k}, \pi_{1k})$ and $n_{21k} \sim \text{bin}(n_{2+k}, \pi_{2k})$ independently, so

$$E(n_{11k}n_{22k}) = n_{1+k}n_{2+k}\pi_{1k}(1-\pi_{2k}), \qquad E(n_{12k}n_{21k}) = n_{1+k}n_{2+k}(1-\pi_{1k})\pi_{2k}.$$

Step 2. Since $\theta = \pi_{1k}(1-\pi_{2k})/\{(1-\pi_{1k})\pi_{2k}\}$, $E(n_{11k}n_{22k}) = \theta E(n_{12k}n_{21k})$, i.e. $E(R_k - \theta S_k) = 0$. Summing gives $E(R - \theta S) = 0$.

Step 3. Under either asymptotic scheme, R/m and S/m (with $m = n$ or $m = K$) are averages of independent terms with bounded variances relative to m, so by the law of large numbers $(R - \theta S)/m \xrightarrow{p} 0$ and S/m stays bounded away from 0 (assuming the strata are nondegenerate). Hence $\hat{\theta}_{MH} - \theta = (R - \theta S)/S \xrightarrow{p} 0$. ■

Theorem 6.21 (Unconditional ML breaks down for matched pairs)

In model (6.2) with one matched pair per stratum, let b, c be as in Proposition 6.19. Then the unconditional ML estimator is $\hat{\beta} = 2\log(b/c)$, whereas $\hat{\theta}_{MH} = b/c$. As $K \to \infty$, $\log(b/c) \xrightarrow{p} \beta$, so $\hat{\beta} \xrightarrow{p} 2\beta$.

Proof. [Supplement]

Step 1. A concordant pair's likelihood is maximized by $\gamma_k = \pm\infty$ and then equals 1, contributing nothing about β.

Step 2. Write $u = e^{\gamma_k}$, $\psi = e^{\beta}$. A pair where only member 1 succeeds has likelihood $\frac{u\psi}{1+u\psi} \cdot \frac{1}{1+u}$, and a pair where only member 2 succeeds has $\frac{1}{1+u\psi} \cdot \frac{u}{1+u}$. Both are proportional in u to $u/\{(1+u\psi)(1+u)\}$, maximized at $u = \psi^{-1/2}$ (derivative of its log: $1/u - \psi/(1+u\psi) - 1/(1+u) = 0 \iff u^2\psi = 1$). The maximized values are $\psi/(1+\psi^{1/2})^2$ and $1/(1+\psi^{1/2})^2$.

Step 3. The profile log likelihood is $b\beta - 2(b+c)\log(1+e^{\beta/2})$, with derivative $b - (b+c)e^{\beta/2}/(1+e^{\beta/2})$, zero at $e^{\beta/2} = b/c$. So $\hat{\beta} = 2\log(b/c)$.

Step 4. For pairs, $R_k = \frac{1}{2}$ in a b-type pair, $S_k = \frac{1}{2}$ in a c-type pair, and both vanish for concordant pairs, so $\hat{\theta}_{MH} = b/c$.

Step 5. Given discordance, a pair is of b-type with probability $\pi_1(1-\pi_2)/\{\pi_1(1-\pi_2) + (1-\pi_1)\pi_2\} = \psi/(1+\psi)$ in every stratum, so $b/(b+c) \xrightarrow{p} \psi/(1+\psi)$ and $b/c \xrightarrow{p} \psi$. ■

Variance of $\log \hat{\theta}_{MH}$

Robins, Breslow and Greenland showed $\log \hat{\theta}_{MH} - \log \theta$ is approximately proportional to $R - \theta S$ and derived an estimated variance valid under both asymptotic schemes:

$$\hat{\sigma}^2(\log \hat{\theta}_{MH}) = \frac{\sum_k (n_{11k} + n_{22k})R_k/n_{++k}}{2R^2} + \frac{\sum_k (n_{12k} + n_{21k})S_k/n_{++k}}{2S^2} + \frac{\sum_k \left[(n_{11k} + n_{22k})S_k + (n_{12k} + n_{21k})R_k\right]/n_{++k}}{2RS}.$$

With fixed K, $\log \hat{\theta}_{MH}$ is slightly less efficient than the ML estimator unless $\beta = 0$; with many sparse strata it is preferred.

Example 6.22 (Clinical trial: common odds ratio)

For Example 6.15,

$$\hat{\theta}_{MH} = \frac{(11 \cdot 27)/73 + \cdots + (4 \cdot 1)/13}{(25 \cdot 10)/73 + \cdots + (2 \cdot 6)/13} = 2.13,$$

$\log \hat{\theta}_{MH} = 0.758$ with $\hat{\sigma} = 0.303$, so the 95% CI is $\exp(0.758 \pm 1.96 \cdot 0.303) = (1.18, 3.87)$. The model-based Wald interval for e^{β} is $(1.19, 3.97)$ and the LR interval $(1.20, 4.02)$. The effect is clear but its size is imprecise.

$\hat{\theta}_{MH}$ remains a useful summary, and CMH a powerful test, when the odds ratios vary moderately but mostly in one direction; equality of odds ratios is not required for either.

6.3.6 Testing homogeneity of odds ratios

Homogeneity $\theta_{XY(1)} = \cdots = \theta_{XY(K)}$ is equivalent to model (6.2), so its G^2 and X^2 goodness-of-fit statistics test it with $df = K - 1$ (the $K - 1$ omitted $X \times Z$ interaction terms).

Proposition 6.23 (Fitted counts under a common odds ratio)

Fix the margins of stratum k and a value $\theta > 0$. There is a unique m_k with $\max(0, n_{1+k} + n_{+1k} - n_{++k}) < m_k < \min(n_{1+k}, n_{+1k})$ such that the table with $(1,1)$ entry m_k and the given margins has odds ratio θ:

$$\frac{m_k\,(n_{2+k} - n_{+1k} + m_k)}{(n_{1+k} - m_k)(n_{+1k} - m_k)} = \theta.$$

Proof. [Supplement] The other three cells are $n_{1+k} - m$, $n_{+1k} - m$ and $n_{2+k} - n_{+1k} + m$, all positive exactly on the open interval stated. On that interval the left side is continuous and strictly increasing in m (numerator increasing, denominator decreasing), tends to 0 at the lower end (a numerator factor vanishes) and to $+\infty$ at the upper end (a denominator factor vanishes). By the intermediate value theorem there is exactly one root; it solves a quadratic equation in m. ■

Definition (Breslow–Day statistic)

With m_k from Proposition 6.23 at $\theta = \hat{\theta}_{MH}$ and v_k the variance of n_{11k} under the fitted table (approximately $[1/m_k + 1/(n_{1+k} - m_k) + 1/(n_{+1k} - m_k) + 1/(n_{2+k} - n_{+1k} + m_k)]^{-1}$),

$$\mathrm{BD} = \sum_k \frac{(n_{11k} - m_k)^2}{v_k},$$

approximately χ^2_{K-1} under homogeneity.

Example 6.24 (Clinical trial: homogeneity)

For Example 6.15, $G^2 = 9.7$ and $X^2 = 8.0$ ($df = 7$): no evidence against a common odds ratio, so $\hat{\theta}_{MH} = 2.1$ summarizes all eight centers. A homogeneity test is not a prerequisite for using $\hat{\theta}_{MH}$ or CMH.

6.3.7 Summarizing heterogeneity in odds ratios

Exactly equal odds ratios are unrealistic (unmeasured covariates vary across strata), and the sampled strata (centers) usually represent a larger population. A *random-effects* logit model takes the stratum log odds ratios as a sample from $N(\mu, \sigma^2)$, estimating a mean effect and its variability; inference then extends to the population of strata (Chapter 12).

6.4 Using models to improve inferential power

Motivation. Tests aimed at a narrow, plausible alternative (such as a linear trend) can be far more powerful than omnibus tests. The noncentral chi-squared distribution explains why: power is gained by concentrating the noncentrality on few degrees of freedom.

6.4.1 Directed alternatives

For an $I \times 2$ table with binomial parameters $\{\pi_i\}$, independence is $\operatorname{logit} \pi_i = \alpha$. The ordinary statistics $X^2(I)$ and $G^2(I)$ test it against the saturated model $\operatorname{logit} \pi_i = \alpha + \beta_i$ ($df = I - 1$), treating rows as nominal. With ordered rows, the linear logit model $\operatorname{logit} \pi_i = \alpha + \beta x_i$ gives the directed test of $\beta = 0$ with statistic

$$G^2(I \mid L) = G^2(I) - G^2(L), \qquad df = 1.$$

A large value on $df = 1$ lies farther in the tail than a comparable value on $df = I - 1$.

6.4.2 Noncentral chi-squared distribution

Definition (Noncentral chi-squared)

If $Z_i \sim N(\mu_i, 1)$, $i = 1, \ldots, \nu$, are independent, then $\sum Z_i^2$ has the noncentral chi-squared distribution $\chi^2_{\nu,\lambda}$ with $df = \nu$ and noncentrality $\lambda = \sum \mu_i^2$. The central case is $\lambda = 0$.

Proposition 6.25 (Moments and monotonicity of power)

(a) $E(\chi^2_{\nu,\lambda}) = \nu + \lambda$ and $\operatorname{Var}(\chi^2_{\nu,\lambda}) = 2(\nu + 2\lambda)$; the distribution depends on the μ_i only through λ.
(b) For fixed ν and critical value $c > 0$, $P(\chi^2_{\nu,\lambda} > c)$ is strictly increasing in λ, equal to α at $\lambda = 0$ when $c = \chi^2_\nu(\alpha)$.

Proof. [Supplement]

Step 1. $E(Z_i^2) = 1 + \mu_i^2$. With $Z_i = \mu_i + \varepsilon_i$, $Z_i^2 = \mu_i^2 + 2\mu_i\varepsilon_i + \varepsilon_i^2$ and $\operatorname{Var}(Z_i^2) = 4\mu_i^2 + 2$ (using $E\varepsilon^3 = 0$, $\operatorname{Var}\varepsilon^2 = 2$). Summing gives (a)'s moments.

Step 2. An orthogonal matrix Q with first row $\mu^\top/\|\mu\|$ maps $Z \sim N(\mu, I)$ to $QZ \sim N((\sqrt{\lambda}, 0, \ldots, 0)^\top, I)$ with $\|QZ\| = \|Z\|$. Hence $\sum Z_i^2 \overset{d}{=} (\varepsilon_1 + \sqrt{\lambda})^2 + \chi^2_{\nu-1}$, depending only on λ.

Step 3. For $a \geq 0$ and $s > 0$, $g(a) = P(|\varepsilon + a| > s) = 1 - \Phi(s - a) + \Phi(-s - a)$ has $g'(a) = \phi(s - a) - \phi(s + a) > 0$ for $a > 0$. Conditioning on the independent $\chi^2_{\nu-1} = w$ (with $s = \sqrt{c - w}$ when $w < c$; the event is certain when $w \geq c$) and integrating shows that $P(\chi^2_{\nu,\lambda} > c)$ is strictly increasing in $a = \sqrt{\lambda}$. ∎

Power decreases with *df*

A fundamental further fact (Das Gupta and Perlman): for fixed λ and α, $P(\chi^2_{\nu,\lambda} > \chi^2_\nu(\alpha))$ decreases as ν increases. So concentrating the noncentrality on a statistic with small *df* increases power.

6.4.3 Increased power for narrower alternatives

Proposition 6.26 (Partitioning G^2)

For the nested models independence $\subset$ linear logit $\subset$ saturated,

$$G^2(I) = G^2(I \mid L) + G^2(L),$$

with df $(I-1) = 1 + (I-2)$.

Proof. With L_0, L_1, L_S the maximized log likelihoods, $G^2(I) = -2(L_0 - L_S) = -2(L_0 - L_1) - 2(L_1 - L_S) = G^2(I \mid L) + G^2(L)$. The df are differences in numbers of parameters: $(I-1) = (2-1) + (I-2)$. ■

When the linear logit model holds, $G^2(L)$ is asymptotically central χ^2_{I-2}, so $G^2(I)$ and $G^2(I \mid L)$ have (approximately) the same noncentrality but df $I-1$ versus 1: the directed test is more powerful. If the linear model fails, $G^2(I)$ has larger noncentrality, but as long as the linear model approximates reality reasonably, the $df = 1$ test still wins. For instance, for $\alpha = 0.05$ a $df = 1$ test beats a $df = 7$ test whenever its noncentrality is at least about half as large. As Mantel noted, testing a linear component does not assume linearity: it provides power against any progressive association. The same reasoning favors the $df = 1$ test of $\beta = 0$ in (6.2) over the $df = K$ goodness-of-fit test of (6.3) when (6.2) holds.

6.4.4 Treatment of leprosy example

Example 6.27 (Leprosy)

Change in clinical condition after 48 weeks of treatment, by degree of infiltration at the start:

Clinical change (score)	High	Low	Prop. high
Worse (−1)	1	11	0.08
Stationary (0)	13	53	0.20
Slight improvement (1)	16	42	0.28
Moderate improvement (2)	15	27	0.36
Marked improvement (3)	7	11	0.39

Comparing mean change between infiltration levels is equivalent to a trend test with infiltration as binary response. $G^2(I) = 7.28$ ($df = 4$, $P = 0.12$) ignores the ordering; the linear logit test gives $G^2(I \mid L) = 6.65$ ($df = 1$, $P = 0.01$), and $G^2(L) = 0.63$ ($df = 3$) shows a good linear fit.

6.4.5 Model smoothing improves precision of estimation

Theorem 6.28 (Model-based estimators are more efficient)

Let p be the sample proportions from a multinomial(n, π) sample with $\pi = \pi(\theta)$ satisfying a smooth model with full-rank $A = \partial\pi/\partial\theta^\top$, and let $\hat{\pi} = \pi(\hat{\theta})$ be the ML fit. For every differentiable g,

$$\text{asymp. Var}\big[\sqrt{n}\, g(\hat{\pi})\big] \le \text{asymp. Var}\big[\sqrt{n}\, g(p)\big].$$

Proof. [Supplement] Let $D = \text{diag}(\pi)$ and $c = \nabla g(\pi)$. By the delta method it suffices to show $c^\top \Sigma_{\hat{\pi}} c \le c^\top \Sigma_p c$ for the asymptotic covariances.

Step 1. $\Sigma_p = D - \pi\pi^\top$. The Fisher information per observation is $A^\top D^{-1} A$, so $\Sigma_{\hat\theta} = (A^\top D^{-1} A)^{-1}$ and $\Sigma_{\hat\pi} = A(A^\top D^{-1} A)^{-1} A^\top$.

Step 2. Put $M = D^{-1/2} A$ and $q = D^{-1/2}\pi = (\sqrt{\pi_i})$, a unit vector. Then $\Sigma_{\hat\pi} = D^{1/2} P_M D^{1/2}$ with $P_M = M(M^\top M)^{-1} M^\top$, and $\Sigma_p = D^{1/2}(I - qq^\top)D^{1/2}$.

Step 3. $M^\top q = A^\top \mathbf{1} = \partial(\sum_i \pi_i)/\partial\theta = 0$, so $q \perp \mathrm{col}(M)$. Then $P_M + qq^\top$ is an orthogonal projection, hence $P_M + qq^\top \le I$, i.e. $P_M \le I - qq^\top$. Sandwiching by $D^{1/2}$ gives $\Sigma_{\hat\pi} \le \Sigma_p$ in the Loewner order. ∎

Even when the model is only approximately true, $\hat\pi_i$ is biased but less variable, and $\mathrm{MSE}(\hat\pi_i) < \mathrm{MSE}(p_i)$ unless n is very large.

6.5 Sample size and power considerations*

Motivation. A study should be large enough to detect effects of practical size with reasonable probability. Normal approximations and the noncentral chi-squared give simple planning formulas.

6.5.1 Sample size and power for comparing two proportions

Proposition 6.29 (Power and sample size for two proportions)

With independent samples of size $n_1 = n_2$ and $\sigma^2 = \pi_1(1-\pi_1) + \pi_2(1-\pi_2)$, the two-sided level-$\alpha$ z test of $\pi_1 = \pi_2$ has approximate power

$$1 - \Phi\Big(z_{\alpha/2} - \frac{|\pi_1 - \pi_2|\sqrt{n_1}}{\sigma}\Big) + \Phi\Big(-z_{\alpha/2} - \frac{|\pi_1 - \pi_2|\sqrt{n_1}}{\sigma}\Big),$$

and power $1-\beta$ requires approximately

$$n_1 = n_2 = \frac{(z_{\alpha/2} + z_\beta)^2\, \sigma^2}{(\pi_1 - \pi_2)^2}.$$

Proof. [Supplement] Let $\Delta = \pi_1 - \pi_2 > 0$ (w.l.o.g.) and $s = \sigma/\sqrt{n_1}$, the standard deviation of $\hat\pi_1 - \hat\pi_2$; take the same s (evaluated near the common value) under H_0. Then $(\hat\pi_1 - \hat\pi_2 - \Delta)/s \approx Z \sim N(0,1)$, and the rejection event $|\hat\pi_1 - \hat\pi_2| \ge z_{\alpha/2} s$ has probability

$$P(Z \ge z_{\alpha/2} - \Delta/s) + P(Z \le -z_{\alpha/2} - \Delta/s),$$

the stated power. Neglecting the second (tiny) term and setting the first equal to $1-\beta$ gives $z_{\alpha/2} - \Delta/s = -z_\beta$, i.e. $\sqrt{n_1} = (z_{\alpha/2} + z_\beta)\sigma/\Delta$. ∎

Example 6.30 (Two treatments with $\pi \approx 0.6$)

With π_1, π_2 near 0.6 and total n ($n/2$ per group), $\mathrm{Var}(\hat\pi_1 - \hat\pi_2) \approx 0.96/n$. For a difference of 0.10 at $\alpha = 0.05$, the power is

$$1 - \Phi(1.96 - 0.10\sqrt{n/0.96}) + \Phi(-1.96 - 0.10\sqrt{n/0.96}),$$

only 0.11 for $n = 50$ and 0.30 for $n = 200$. For $\pi_1 = 0.6$, $\pi_2 = 0.7$, $\alpha = 0.05$, $\beta = 0.10$:

$$n_1 = n_2 = (1.96 + 1.28)^2(0.24 + 0.21)/0.01 \approx 473.$$

The formula may slightly underestimate the required size.

6.5.2 Sample size determination in logistic regression

Hsieh's formula

For logit $\pi(x) = \beta_0 + \beta_1 x$ with x approximately normal, let $\bar{\pi}$ be the probability at the mean of x and $\tau = \beta_1 s_x$ the log odds ratio for a one-standard-deviation increase. A one-sided level-α test of $\beta_1 = 0$ with power $1 - \beta$ needs approximately

$$n = \frac{\left[z_\alpha + z_\beta \exp(-\tau^2/4)\right]^2 (1 + 2\bar{\pi}\delta)}{\bar{\pi}\,\tau^2}, \qquad \delta = \frac{1 + (1+\tau^2)\exp(5\tau^2/4)}{1 + \exp(-\tau^2/4)}.$$

n decreases as $\bar{\pi} \to 0.5$ and as $|\tau|$ grows.

Example 6.31 (Cholesterol and heart disease)

$\bar{\pi} = 0.08$; a one-SD cholesterol increase should raise it to 0.12. Odds: $0.087 \to 0.136$, odds ratio 1.57, $\tau = 0.450$. For $\alpha = 0.05$, $\beta = 0.10$: $\delta = 1.306$ and $n = 612$.

6.5.3 Sample size in multiple logistic regression

If R is the multiple correlation between the predictor of interest and the other predictors, divide n by $1 - R^2$ (with $\bar{\pi}$ evaluated at the means). With blood pressure correlated 0.40 with cholesterol, $n \approx 612/(1 - 0.16) = 729$. These are rough guides only.

6.5.4 Power for chi-squared tests in contingency tables

Let H_0 correspond to model M, let π_i be the true cell probabilities and $\pi_i(M)$ the limit of the ML fit of M.

Definition (Noncentrality for X^2 and G^2)

For a multinomial sample of size n,

$$\lambda_{X^2} = n \sum_i \frac{[\pi_i - \pi_i(M)]^2}{\pi_i(M)}, \qquad \lambda_{G^2} = 2n \sum_i \pi_i \log \frac{\pi_i}{\pi_i(M)},$$

i.e. the statistics evaluated with π in place of p and $\pi(M)$ in place of $\hat{\pi}$. Under H_0, $\lambda = 0$.

Power is computed by:

(1) choosing hypothetical $\{\pi_i\}$;

(2) fitting M to them to get $\{\pi_i(M)\}$;

(3) computing λ;

(4) evaluating $P(\chi^2_{\nu,\lambda} > \chi^2_\nu(\alpha))$. Excerpt of power at $\alpha = 0.05$:

	Noncentrality λ						
df	1.0	2.0	3.0	5.0	7.0	10.0	15.0
1	.170	.293	.410	.609	.754	.885	.972
2	.133	.226	.322	.504	.655	.815	.944
4	.106	.172	.244	.396	.540	.716	.891
10	.082	.121	.166	.268	.379	.542	.760

6.5.5 Power for testing conditional independence

Example 6.32 (Fetal heart rate monitoring)

Response: nonroutine care needed in the first week. Does a new test N (very/somewhat worrisome, reassuring) predict it given the standard test S (worrisome, reassuring)? The physician's scenario:

Standard	New	Joint prob.	P(nonroutine care)
Worrisome	Very worrisome	0.04	0.40
	Somewhat worrisome	0.08	0.32
	Reassuring	0.04	0.27
Reassuring	Very worrisome	0.02	0.30
	Somewhat worrisome	0.18	0.22
	Reassuring	0.64	0.15

Products give $\{\pi_{ijk}\}$ (e.g. $0.04 \times 0.40 = 0.016$). Fitting the logit models with $N + S$ and with S only to these "data" gives $\pi(M_1)$, $\pi(M_0)$, and the LR noncentrality is $0.00816\,n$ with $df = 2$. Powers at $\alpha = 0.05$ are 0.35, 0.49, 0.73 for $n = 400, 600, 1000$; the concentration of 64% of cases in one cell weakens power.

6.5.6 Effects of sample size on model selection and inference

Small samples accept very simple models; large samples reject all but complex ones, and some significant effects are substantively trivial. With large n a model simpler than those passing goodness-of-fit tests may suffice. Always estimate effects: a confidence interval shows whether a rejected H_0 is wrong by an important amount, and whether a nonrejected H_0 still leaves important alternatives plausible.

6.6 Probit and complementary log-log models*

Motivation. The logit is not the only sensible link for a binary response. Tolerance-distribution arguments lead to links of the form $\pi(x) = \Phi(\alpha + \beta x)$ for any continuous cdf Φ, giving probit and (asymmetric) complementary log-log models.

These models have the form

$$\pi(x) = \Phi(\alpha + \beta x) \tag{6.6}$$

for a continuous cdf Φ.

6.6.1 Tolerance motivation for binary response models

Proposition 6.33 (Tolerance distributions give cdf links)

Suppose a subject responds ($Y = 1$, e.g. dies) at dosage x iff its tolerance $T \le x$, where T has cdf $F(t) = \Phi((t - \mu)/\sigma)$ for a standard cdf Φ. Then $\pi(x) = \Phi(\alpha + \beta x)$ with $\alpha = -\mu/\sigma$ and $\beta = 1/\sigma$.

Proof. $\pi(x) = P(Y = 1 \mid x) = P(T \le x) = F(x) = \Phi\big((x - \mu)/\sigma\big) = \Phi(-\mu/\sigma + x/\sigma)$. ■

6.6.2 Probit models

Definition (Probit model)

With Φ the standard normal cdf (normal tolerances for, e.g., log concentration),

$$\Phi^{-1}[\pi(x)] = \alpha + \beta x.$$

Proposition 6.34 (Comparing probit and logit slopes)

(a) For the probit model the curve is a $N(-\alpha/\beta,\ 1/\beta^2)$ cdf (for $\beta > 0$); it rises from $\pi = 0.16$ to 0.50 over a distance $1/|\beta|$, and its steepest slope, at $\pi = \frac{1}{2}$, is $\beta\phi(0) = 0.40\beta$.
(b) For the logit model with slope β the curve is a logistic cdf with standard deviation $\pi/(\sqrt{3}|\beta|)$ and steepest slope 0.25β.
(c) Hence the two curves have equal slopes at $\pi = \frac{1}{2}$ when $\beta_{\text{logit}} = 1.6\,\beta_{\text{probit}}$, and equal standard deviations when $\beta_{\text{logit}} = (\pi/\sqrt{3})\beta_{\text{probit}} \approx 1.8\,\beta_{\text{probit}}$.

Proof.

(a) $\Phi(\alpha+\beta x) = \Phi\big((x-\mu)/\sigma\big)$ with $\mu = -\alpha/\beta$, $\sigma = 1/\beta$. Since $\Phi(-1) \approx 0.16$, $\pi = 0.16$ at $x = \mu - \sigma$. The derivative is $\beta\phi(\alpha + \beta x)$, maximal where $\alpha + \beta x = 0$, equal to $\beta/\sqrt{2\pi} \approx 0.40\beta$.

(b) The standard logistic cdf $F(z) = e^z/(1 + e^z)$ has density $F(1 - F)$, maximal $1/4$ at $z = 0$, and variance $\pi^2/3$; so $F(\alpha + \beta x)$ has slope at most 0.25β and is the cdf of a variable with standard deviation $\pi/(\sqrt{3}|\beta|)$.

(c) Equate $0.25\beta_{\text{logit}} = 0.40\beta_{\text{probit}}$, and $\pi/(\sqrt{3}\beta_{\text{logit}}) = 1/\beta_{\text{probit}}$. ■

Thus logit estimates are typically 1.6–1.8 times probit estimates. Fisher scoring (expected information) and Newton–Raphson (observed information) give the same ML estimates but slightly different SEs for noncanonical links such as the probit.

6.6.3 Beetle mortality example

Example 6.35 (Beetle mortality)

Beetles killed after 5 hours of exposure to gaseous carbon disulfide:

			Fitted values		
Log dose	Beetles	Killed	C. log-log	Probit	Logit
1.691	59	6	5.7	3.4	3.5
1.724	60	13	11.3	10.7	9.8
1.755	62	18	20.9	23.4	22.4
1.784	56	28	30.3	33.8	33.9
1.811	63	52	47.7	49.6	50.0
1.837	59	53	54.2	53.4	53.3
1.861	62	61	61.1	59.7	59.2
1.884	60	60	59.9	59.2	58.8

Probit fit: $\Phi^{-1}[\hat{\pi}(x)] = -34.96 + 19.74x$; $\hat{\pi} = 0.5$ at $x = 34.96/19.74 = 1.77$, i.e. a normal tolerance distribution with $\mu = 1.77$, $\sigma = 1/19.74 = 0.05$. Probit and logit fit similarly and rather poorly: $G^2 = 10.0$ and 11.1 ($df = 6$). The observed proportions rise slowly and then approach 1 sharply, an asymmetric pattern.

6.6.4 Complementary log-log link models

A link is *symmetric* if $\text{link}[\pi] = -\text{link}[1-\pi]$; e.g. $\text{logit}(\pi) = \log\frac{\pi}{1-\pi} = -\log\frac{1-\pi}{\pi} = -\text{logit}(1-\pi)$, and $\Phi^{-1}(\pi) = -\Phi^{-1}(1-\pi)$. Then $\pi(x)$ approaches 0 and 1 at the same rate.

Definition (Complementary log-log and log-log models)

$$\pi(x) = 1 - \exp[-\exp(\alpha+\beta x)], \qquad \log\{-\log[1-\pi(x)]\} = \alpha+\beta x;$$
$$\pi(x) = \exp[-\exp(\alpha+\beta x)], \qquad \log\{-\log[\pi(x)]\} = \alpha+\beta x.$$

The cloglog curve approaches 0 slowly and 1 sharply; the log-log curve the reverse. If the cloglog model holds for success, the log-log model (with $-\beta$, $-\alpha$) holds for failure.

Proposition 6.36 (Interpretation and latent extreme-value distributions)

(a) Under the cloglog model, for any x_1, x_2,

$$1-\pi(x_2) = [1-\pi(x_1)]^{\exp[\beta(x_2-x_1)]};$$

a unit increase in x raises the complement probability to the power e^{β}.
(b) The log-log model is (6.6) with Φ the Gumbel (largest extreme value) cdf $F(x) = \exp\{-\exp[-(x-a)/b]\}$, $b>0$ (mean $a+0.577b$, sd $\pi b/\sqrt{6}$), with $\beta<0$ for π increasing in x; the cloglog model corresponds to a tolerance with the smallest-extreme-value cdf $F(t) = 1-\exp\{-\exp[(t-a)/b]\}$.

Proof.

(a) $\log[-\log(1-\pi(x_2))] - \log[-\log(1-\pi(x_1))] = \beta(x_2-x_1)$, so $\log(1-\pi(x_2)) = e^{\beta(x_2-x_1)}\log(1-\pi(x_1))$; exponentiate.

(b) $\exp\{-\exp[\alpha+\beta x]\}$ equals the Gumbel cdf $\exp\{-\exp[-(x-a)/b]\}$ evaluated at x when $\beta = -1/b$ and $\alpha = a/b$. For the cloglog model, the tolerance argument (Proposition 6.33) with $F(t) = 1-\exp\{-\exp[(t-a)/b]\}$ gives $\pi(x) = 1-\exp\{-\exp(-a/b+x/b)\}$. ■

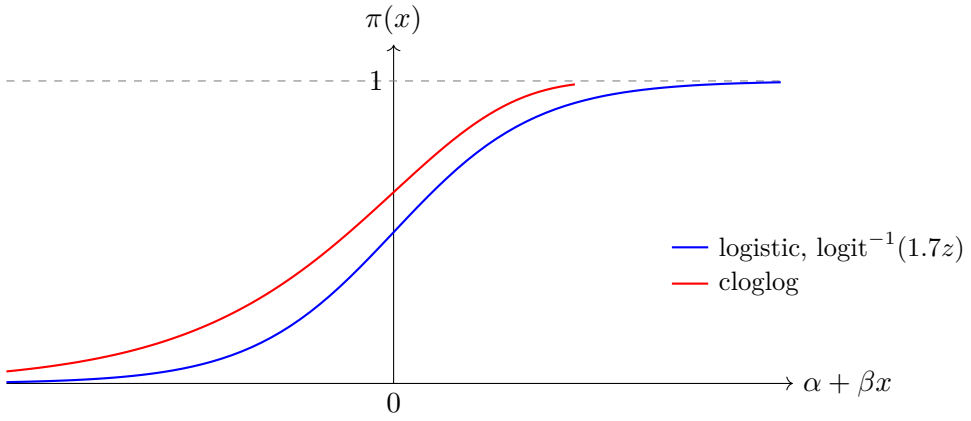


Figure 1: Symmetric (logistic, scaled to resemble the probit) versus asymmetric complementary log-log response curves: the cloglog curve leaves 0 slowly and reaches 1 sharply.

6.6.5 Beetle mortality example revisited

Example 6.37 (Beetles: cloglog fit)

The cloglog model gives $\hat{\alpha} = -39.52$, $\hat{\beta} = 22.01$. The estimated survival probability $1-\hat{\pi}(x) = \exp[-\exp(-39.52 + 22.01x)]$ is 0.885 at $x = 1.7$, 0.332 at $x = 1.8$, and 5×10^{-5} at $x = 1.9$. Survival at $x + 0.1$ equals survival at x raised to the power $\exp(22.01 \times 0.1) = 9.03$; e.g. $0.885^{9.03} = 0.332$. The fit is good ($G^2 = 3.5$, $df = 6$).

6.7 Conditional logistic regression and exact distributions*

Motivation. When n is small, or the number of nuisance parameters grows with n, ordinary ML inference fails. Conditioning on the sufficient statistics of nuisance parameters removes them from the likelihood and yields exact small-sample inference, generalizing Fisher's exact test.

6.7.1 Conditional likelihood

Let $y_i \in \{0, 1\}$, $i = 1, \ldots, N$ (single trials), with predictors x_{ij}, and

$$P(Y_i = y_i) = \frac{\exp\left[y_i(\alpha + \sum_{j=1}^{p} \beta_j x_{ij})\right]}{1 + \exp(\alpha + \sum_{j=1}^{p} \beta_j x_{ij})}. \tag{6.7}$$

For independent observations

$$P(Y_1 = y_1, \ldots, Y_N = y_N) = \frac{\exp\left[\alpha \sum_i y_i + \sum_j \beta_j \sum_i y_i x_{ij}\right]}{\prod_i [1 + \exp(\alpha + \sum_j \beta_j x_{ij})]},$$

so the sufficient statistics are $\sum_i y_i$ for α and $T_j = \sum_i y_i x_{ij}$ for β_j.

Theorem 6.38 (Conditioning eliminates the intercept)

Let $S(t) = \{(y_1^*, \ldots, y_N^*) \in \{0,1\}^N : \sum_i y_i^* = t\}$. For observed y with $\sum_i y_i = t$,

$$P\Big(Y = y \;\Big|\; \textstyle\sum_i Y_i = t\Big) = \frac{\exp\left(\sum_j \beta_j \sum_i y_i x_{ij}\right)}{\sum_{y^* \in S(t)} \exp\left(\sum_j \beta_j \sum_i y_i^* x_{ij}\right)},$$

which does not depend on α.

Proof. $P(Y = y \mid \sum Y_i = t) = P(Y = y) / \sum_{y^* \in S(t)} P(Y = y^*)$. In each probability the denominator $\prod_i [1 + \exp(\alpha + \sum_j \beta_j x_{ij})]$ is the same (it does not involve y) and the factor $e^{\alpha t}$ is common because every $y^* \in S(t)$ has $\sum_i y_i^* = t$. Both cancel, leaving the stated ratio. ■

The conditional likelihood is used like an ordinary one: its maximizers (conditional ML estimates) are asymptotically normal with covariance the inverse of the negative Hessian of the conditional log likelihood.

6.7.2 Small-sample conditional inference for logistic regression

Theorem 6.39 (Exact conditional distribution of a sufficient statistic)

Let $x_{i0} = 1$, $T_j = \sum_i y_i x_{ij}$, and condition on $T_j = t_j$, $j = 0, \dots, p-1$. Let $c(t_0, \dots, t_{p-1}, u)$ be the number of $y^* \in \{0,1\}^N$ with $\sum_i y_i^* x_{ij} = t_j$ $(j < p)$ and $\sum_i y_i^* x_{ip} = u$. Then

$$P(T_p = t \mid T_j = t_j,\ j < p) = \frac{c(t_0, \dots, t_{p-1}, t)\, e^{\beta_p t}}{\sum_u c(t_0, \dots, t_{p-1}, u)\, e^{\beta_p u}}, \tag{6.8}$$

depending only on β_p. For $H_0 : \beta_p = 0$ against $\beta_p > 0$, the exact P-value is

$$\frac{\sum_{t \ge t_{\text{obs}}} c(t_0, \dots, t_{p-1}, t)}{\sum_u c(t_0, \dots, t_{p-1}, u)},$$

the proportion of data configurations in the conditional reference set whose T_p is at least as large as observed.

Proof.

Step 1. As in Theorem 6.38, $P(Y = y^*) \propto \exp(\sum_{j=0}^{p} \beta_j \sum_i y_i^* x_{ij})$ with a normalizing constant free of y^* (here $\beta_0 = \alpha$).

Step 2. On the set $S(t_0, \dots, t_{p-1})$ of y^* with $\sum_i y_i^* x_{ij} = t_j$ for $j < p$, the factor $\exp(\sum_{j<p} \beta_j t_j)$ is constant, so $P(Y = y^* \mid S) \propto \exp(\beta_p \sum_i y_i^* x_{ip})$.

Step 3. Summing over the $c(t_0, \dots, t_{p-1}, u)$ configurations with $T_p = u$ gives (6.8). At $\beta_p = 0$ every configuration in the reference set is equally likely, which gives the P-value. ∎

Computation of the counts $c(\cdot)$ is intensive and needs specialized software (LogXact, SAS `PROC LOGISTIC` with `EXACT`, R package `elrm` or `logistiX`).

6.7.3 Small-sample conditional inference for 2×2 contingency tables

Corollary 6.40 (Fisher's exact test)

For logit $P(Y_i = 1) = \alpha + \beta x_i$ with $x_i \in \{0, 1\}$ (row 1: $x = 1$, n_1 subjects; row 2: n_2 subjects), let s_1, s_2 be the success counts. Given $s = s_1 + s_2$,

$$P(s_1 = t \mid s) = \frac{\binom{n_1}{t}\binom{n_2}{s-t} e^{\beta t}}{\sum_u \binom{n_1}{u}\binom{n_2}{s-u} e^{\beta u}},$$

the noncentral hypergeometric distribution with odds ratio e^{β}. The exact test of $\beta = 0$ is Fisher's exact test.

Proof. Here $T_0 = \sum_i y_i = s$ and $T_1 = \sum_i y_i x_i = s_1$. The number of 0/1 vectors with $s_1 = t$ successes among the n_1 row-1 subjects and $s - t$ among the n_2 row-2 subjects is $c(s, t) = \binom{n_1}{t}\binom{n_2}{s-t}$. Substitute in (6.8). At $\beta = 0$ this is the hypergeometric distribution of Fisher's test. ∎

6.7.4 Small-sample conditional inference for the linear logit model

For an $I \times 2$ table with $y_i \sim \text{bin}(n_i, \pi_i)$ and $\text{logit}\,\pi_i = \alpha + \beta x_i$, conditioning on $s = \sum_i y_i$ removes α and the sufficient statistic $T = \sum_i x_i y_i$ has

$$P(T = t \mid s) = \frac{c(s,t)e^{\beta t}}{\sum_u c(s,u)e^{\beta u}}, \qquad c(s,u) = \sum_{\substack{(y_1,\ldots,y_I):\ \sum y_i = s \\ \sum x_i y_i = u}} \prod_i \binom{n_i}{y_i},$$

by the argument of Theorem 6.39 applied to grouped data. At $\beta = 0$ the counts are multiple hypergeometric, and ordering tables by T is the same as ordering them by the Cochran–Armitage statistic: this is an *exact trend test*. For the (very unbalanced) maternal alcohol and malformation data ($n = 32{,}573$), the exact one-sided P-value is 0.0168 and the two-sided 0.0172, close to the large-sample 0.010.

6.7.5 Small-sample tests of conditional independence in $2 \times 2 \times K$ tables

Theorem 6.41 (Conditional distribution in $2 \times 2 \times K$ tables)

In model (6.2) with $\theta = e^{\beta}$, given all row totals n_{i+k} and column totals n_{+jk},

$$P(n_{11k} = t_k,\ k = 1,\ldots,K) = \prod_{k=1}^{K} \frac{\binom{n_{1+k}}{t_k}\binom{n_{++k}-n_{1+k}}{n_{+1k}-t_k}\theta^{t_k}}{\sum_u \binom{n_{1+k}}{u}\binom{n_{++k}-n_{1+k}}{n_{+1k}-u}\theta^{u}}. \tag{6.9}$$

At $\theta = 1$ it is a product of hypergeometric laws, which determines the exact null distribution of $\sum_k n_{11k}$.

Proof.

Step 1. Strata are independent, and within stratum k the rows are independent binomials with logits $\alpha + \beta + \beta_k^Z$ and $\alpha + \beta_k^Z$. The sufficient statistics are n_{+1k} for the stratum parameters and $\sum_k n_{11k}$ for β; the row totals are fixed by design.

Step 2. Conditioning stratum k on n_{+1k} is Corollary 6.40 with $n_1 = n_{1+k}$, $n_2 = n_{++k} - n_{1+k}$, $s = n_{+1k}$, which gives the kth factor. Independence across strata gives the product. ∎

For $H_a : \beta > 0$ the exact P-value is $P_0(\sum_k n_{11k} \geq \text{observed})$; for $K = 1$ this is Fisher's exact test.

6.7.6 Promotion discrimination example

Example 6.42 (Promotion decisions by race)

U.S. government computer specialists of similar seniority:

	July		August		September	
Race	Yes	No	Yes	No	Yes	No
Black	0	7	0	7	0	8
White	4	16	4	13	2	13

With a zero marginal count after collapsing over month, the CMH asymptotics are doubtful. For $H_a : \beta < 0$ (lower promotion probability for blacks), $\sum_k n_{11k}$ ranges over 0 to 10 and the observed value 0 is the most extreme possible. The exact one-sided P-value is 0.026; the two-sided P-value (summing tables no more likely than observed) is 0.056.

6.7.7 Exact conditional estimation and comparison of odds ratios

The *conditional ML estimator* $\hat{\theta}_{\text{cond}}$ maximizes (6.9) in θ. Since the number of parameters does not grow with K, it behaves well under both standard and sparse asymptotics; it is slightly more efficient than $\hat{\theta}_{MH}$ except at $\theta = 1$ (equally efficient) and for matched pairs (identical, both b/c). Inverting the exact tests of $H_0 : \theta = \theta_0$ based on $\sum_k n_{11k}$ gives a confidence interval with coverage at least the nominal level. For the promotion data $\sum_k n_{11k} = 0$, so the lower bound is 0; the exact 95% interval is $(0, 1.01)$. Zelen proposed an exact test of homogeneity of odds ratios.

6.7.8 Diarrhea example

Example 6.43 (Cephalexin and diarrhea)

2493 hospital patients; predictors age (1 if over 50), length of stay (1 if over a week), Cephalexin exposure:

Cephalexin	Age	Stay	Diarrhea cases	Sample size
0	0	0	0	385
0	0	1	5	233
0	1	0	3	789
0	1	1	47	1081
1	1	1	5	5

All exposed patients had diarrhea, so the ML estimate of the Cephalexin effect in the main-effects model is $+\infty$. Conditioning on the sufficient statistics of the other parameters, the exact 95% CI for the odds ratio is $(19, \infty)$ and $P < 0.0001$ for no effect. Only the older, long-stay subjects carry information about the Cephalexin effect; the first three rows contribute nothing.

6.7.9 Complications from discreteness

Limitations of exact conditional inference

(a) Like Fisher's test, exact conditional tests are conservative because of discreteness, especially for small or unbalanced data. Mid-P-values reduce conservativeness without guaranteed coverage: for the promotion data the mid-P interval is $(0, 0.78)$ rather than $(0, 1.01)$.
(b) If no other y^* shares the observed values of the conditioned sufficient statistics (common when a conditioned-out predictor is continuous with unequally spaced values), the conditional distribution is degenerate and the P-value is 1.
(c) Reduction by sufficiency requires the canonical link: the approach works for logit models but not for probit models.

6.8 R lab

Motivation. Reproduce the chapter's main analyses: model selection by AIC, residual and influence diagnostics, CMH and Mantel–Haenszel inference, exact conditional tests, and probit/cloglog links.

Model selection with `step`. The crab data (`Crabs.dat` from Agresti's web site) has columns `y`, `color`, `spine`, `width`, `weight`; `step` performs backward elimination by AIC, dropping whole factors.

```
crabs <- read.table("Crabs.dat", header = TRUE)
crabs$C <- factor(crabs$color); crabs$S <- factor(crabs$spine)
```

```
full <- glm(y ~ C * S * width, family = binomial, data = crabs)
best <- step(full, direction = "backward")       # AIC-based
crabs$dark <- as.numeric(crabs$color == 4)
fit8 <- glm(y ~ dark + width, family = binomial, data = crabs)
AIC(fit8)                                        # about 194.0
```

Residuals and influence (heart disease, grouped data). `rstandard(type="pearson")` gives the standardized Pearson residuals r_i; `influence.measures` gives Dfbeta-type and Cook-type measures.

```
bp  <- c(111.5,121.5,131.5,141.5,151.5,161.5,176.5,191.5)
n   <- c(156,252,284,271,139,85,99,43)
chd <- c(3,17,12,16,12,8,16,8)
fit <- glm(cbind(chd, n - chd) ~ bp, family = binomial)
c(deviance(fit), sum(residuals(fit, "pearson")^2))   # 5.9, 6.3
round(cbind(pearson = residuals(fit, "pearson"),
            std = rstandard(fit, type = "pearson")), 2)
influence.measures(fit)
```

CMH, Mantel–Haenszel, homogeneity, exact test. `mantelhaen.test` reports CMH (without continuity correction if `correct=FALSE`) and $\hat{\theta}_{MH}$ with its RBG confidence interval; `exact=TRUE` gives the conditional test of Theorem 6.41.

```
cnt <- c(11,10,25,27, 16,22,4,10, 14,7,5,12, 2,1,14,16,
         6,0,11,12, 1,0,10,10, 1,1,4,8, 4,6,2,1)
trial <- array(cnt, dim = c(2,2,8), dimnames = list(
  Trt = c("Drug","Control"), Resp = c("S","F"), Center = 1:8))
mantelhaen.test(trial, correct = FALSE)   # CMH 6.38, MH 2.13 (1.18,3.87)
d <- as.data.frame(as.table(trial))
fit <- glm(Resp == "S" ~ Trt + Center, family = binomial,
           weights = Freq, data = d)       # beta for Control: -0.777
promo <- array(c(0,4,7,16, 0,4,7,13, 0,2,8,13), dim = c(2,2,3))
mantelhaen.test(promo, exact = TRUE, alternative = "less")  # P = 0.026
```

Probit and complementary log-log links (beetles).

```
dose <- c(1.691,1.724,1.755,1.784,1.811,1.837,1.861,1.884)
n    <- c(59,60,62,56,63,59,62,60)
kill <- c(6,13,18,28,52,53,61,60)
links <- c("logit", "probit", "cloglog")
fits <- lapply(links, function(l)
  glm(cbind(kill, n - kill) ~ dose, family = binomial(link = l)))
sapply(fits, deviance)            # 11.1, 10.0, 3.5 (approximately)
coef(fits[[3]])                   # -39.52, 22.01
```